

Yanghao Chen

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Education

Tongji University

Bachelor of Engineering in Engineering Mechanics

GPA: 87/100

Shanghai, China

Sep. 2023 – Present; expected Jun. 2027

Johns Hopkins University

Visiting Undergraduate Research Intern, Department of Civil and Systems Engineering

Baltimore, MD, USA

Jan. 2026 – Oct. 2026

Research Interests: Computational Mechanics, Scientific Machine Learning

Research Projects

GPU-Accelerated Hybrid FEM–Neural Operator Solver

Jan. 2026 – Present

Lead Undergraduate Researcher | Advisor: *Prof. Somdatta Goswami*

Johns Hopkins University

- Developed a hybrid FEM–neural operator solver based on non-overlapping domain decomposition, with FEM and neural-operator subdomains connected through a shared interface.
- Sampled interface displacement fields using Gaussian random fields (GRFs) and used GPU-accelerated JAX-FEM simulations to generate training data. Trained Transolver in JAX to learn mappings from interface displacement fields to subdomain stress fields.
- Computed interface reaction forces from predicted stresses in the neural-operator subdomain and transferred them to the FEM solver. The FEM solver used these forces to compute updated interface displacements and returned them to the neural operator for the next coupling iteration.

PINNs for Shock Capturing in the Burgers Equation

Nov. 2025 – Mar. 2026

Undergraduate Researcher | Advisor: *Prof. Xianyang (Tom) Chen*

Tongji University

- Trained a PINN in PyTorch for the one-dimensional inviscid Burgers shock problem, using PDE-residual, initial-condition, and boundary-condition losses with automatic differentiation.
- Added an artificial-viscosity term with a fixed coefficient to the PDE residual and trained a second PINN to predict the shock profile.
- Compared standard and artificial-viscosity PINN predictions against the analytical inviscid solution, analyzing shock location, transition width, and errors near the discontinuity.

ConvLSTM Prediction of Concentration and Stress Fields in Battery Materials

Jun. 2025 – Oct. 2025

Undergraduate Researcher | Advisor: *Prof. Ying Zhao*

Tongji University

- Generated polycrystalline NMC microstructures and automated MATLAB–COMSOL simulations to produce time-aligned image sequences of lithium concentration and von Mises stress.
- Trained separate conditional ConvLSTM models in PyTorch using past image sequences, grain-orientation maps, and C-rate inputs, with MSE and SSIM losses and scheduled sampling.
- Predicted future concentration and stress image sequences, compared them with COMSOL reference sequences, and analyzed error accumulation during autoregressive prediction.

Domain-Specific LLM Fine-Tuning for Mechanics of Materials

Sep. 2024 – Apr. 2025

Undergraduate Researcher | Advisor: *Prof. Ying Zhao*

Tongji University

- Built instruction–response datasets from textbooks and technical literature, covering stress analysis, constitutive laws, and failure theories.
- Fine-tuned Qwen2.5-7B with LoRA on Google Colab for mechanics of materials Q&A and concept explanations.
- Released two instruction datasets and two fine-tuned Qwen2.5-based checkpoints on [Hugging Face](#).

Technical Skills

Scientific Machine Learning

Computational Mechanics

Programming & Tools

LLM Fine-Tuning

Languages

PyTorch, JAX, Neural Operators (Transolver, DeepONet), PINNs, ConvLSTM
JAX-FEM, FEniCSx, Abaqus, COMSOL Multiphysics (LiveLink for MATLAB), Gmsh
Python, MATLAB, Linux, Git/GitHub, Matplotlib, Origin
Qwen2.5, LoRA, Instruction Dataset Curation
Mandarin Chinese (native), Cantonese, English